



Harsh Priya HW1 - Assignment 1

Analog CMOS Design (Technische Universiteit Delft)



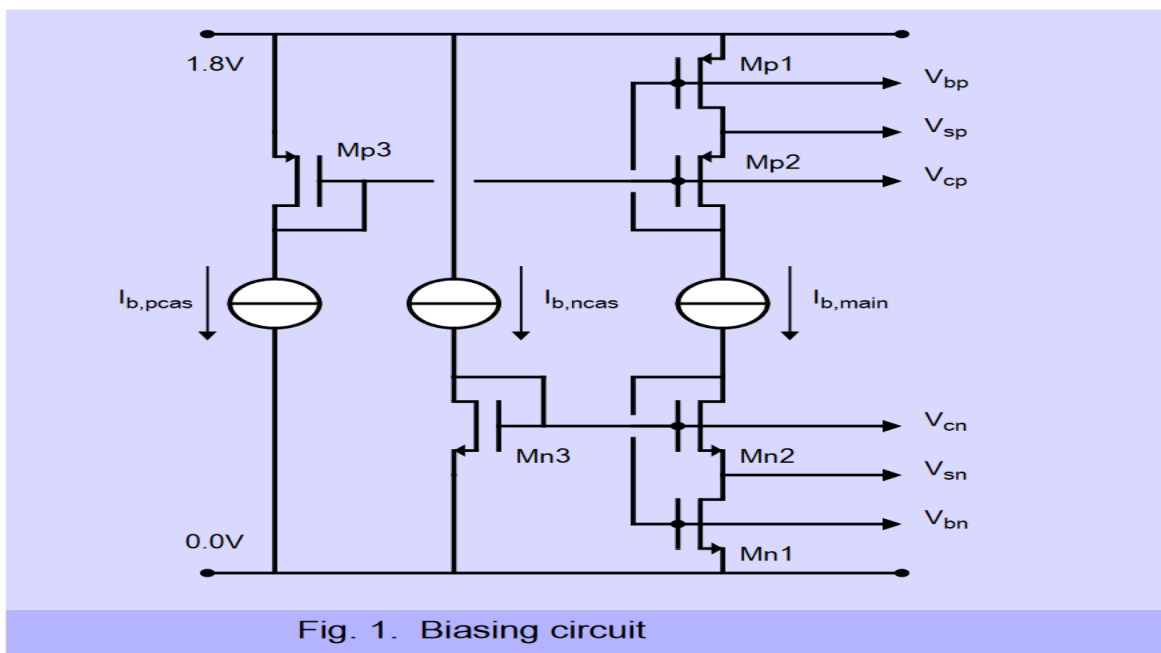
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Analog CMOS Assignment Report – I

Harsh Priya (Student ID – 6194982)

In the assignment, the focus centres around biasing of an amplifier. With the help of cascode current mirror topology, high output impedance and higher output swing can be achieved when compared with a conventional topology. A well designed biased circuit with properly adjusted current ratio ensures that MOSFET(s) operates within its intended region. Even at the margin of saturation, the device must exhibit stable characteristics allowing the amplification of the circuit. The current mirror biasing topology helps in establishing limits of proper operating conditions for active components.

Simulation 1



Given Details: -

- Here, $W=1\mu$ and $L=0.18\mu$ for all the transistors.
- Multiplier factors M_n and M_p should be determined keeping in mind $M_{n1} = M_{n2} = M_{n3} = M_n$ and $M_{p1} = M_{p2} = M_{p3} = M_p$.
- $I_{b,main} = 200\mu A$.

Required Targets to be achieved in Simulation 1: -

- Choosing the values of M_n and M_p such that $V_{gtn1} = V_{gtn2} \approx 100mV$.
- In the problem statement, its given that $I_{b,ncas}$ and $I_{b,pcas}$ should be 4 times larger than $I_{b,main}$. But in practice, the operation points between the triode region and saturation region for all the transistors is not well-bounded. So, the task is to choose ratios of $I_{b,ncas}$ ($ratio_n$) and $I_{b,pcas}$ ($ratio_p$) in such a way that the transistors M_{n1} and M_{p1} are on the verge of saturation with a margin of approximately $50mV$.
- Finding out how the circuit characteristics changes when $I_{b,main}$ values is reduced by 2x and 4x respectively keeping the ratio of $I_{b,ncas} = ratio_n * I_{b,main}$ and $I_{b,pcas} = ratio_p * I_{b,main}$ same throughout both the conditions.

Solutions: -

The final schematic diagram is as follows: -

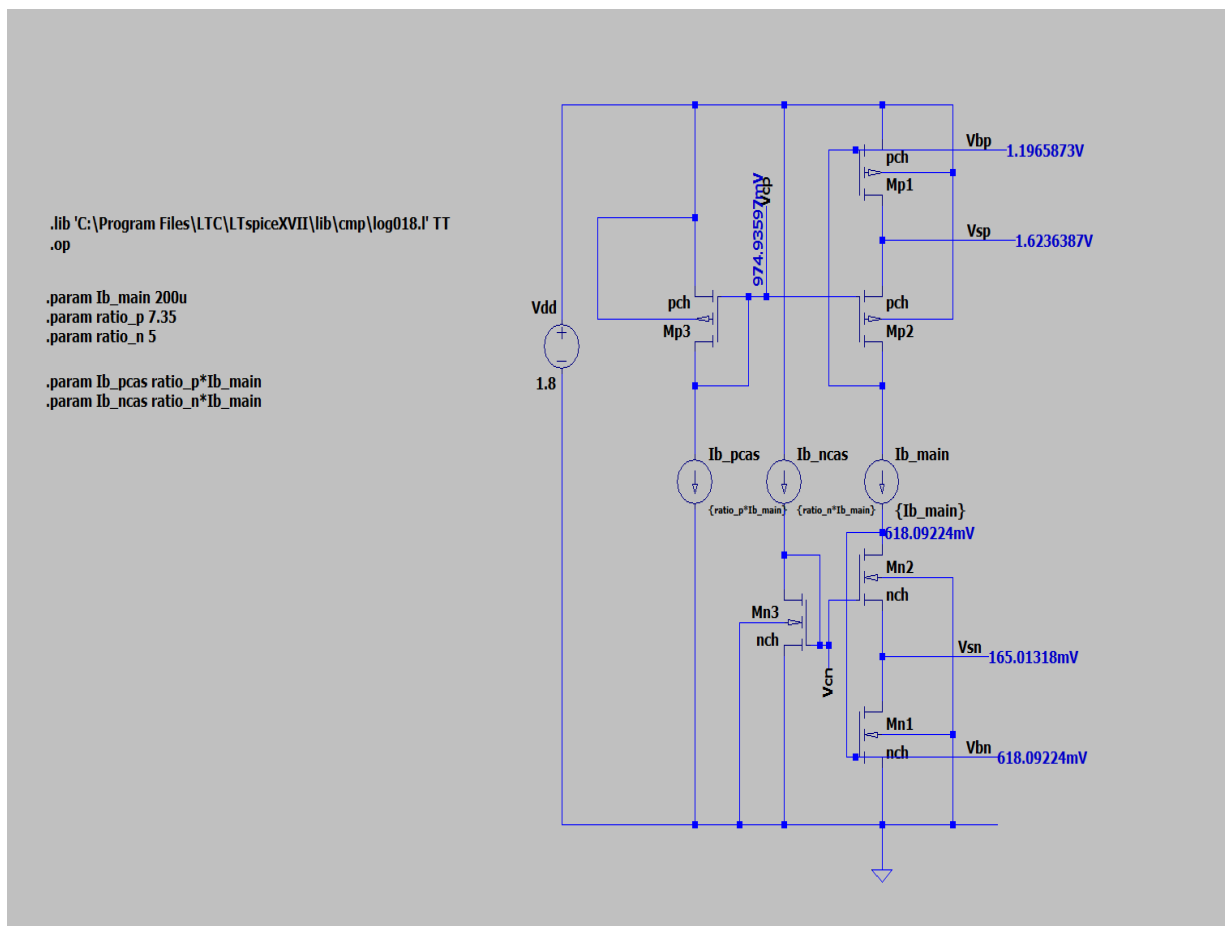


Fig 2. Schematic Diagram for Simulation1 Circuit

(Part a)

The sizing parameters M_n and M_p taken for all the transistors is summarized in Table 1 . as follows: -

	W/L	M(ultiplier) Factor
PMOS	1um/0.18um	58
NMOS	1um/0.18um	11

Table 1. Sizing Parameters Details for PMOS and NMOS

Using these values of M factors, we can verify that the value of V_{gt} for transistors M_{n1} , M_{n2} and M_{p1} , M_{p2} are approximately maintained around $100mV$ using the details in **Spice Error Log** window as given below in Fig 3:

SPICE Error Log: C:\Users\harsh\Downloads\HW1_Sim1.log					
--- BSIM3 MOSFETS ---					
Name:	mn3	mn1	mn2	mp2	mp1
Model:	nch.3	nch.3	nch.3	pch.3	pch.3
Id:	1.00e-03	2.00e-04	2.00e-04	-2.00e-04	-2.00e-04
Vgs:	8.12e-01	6.18e-01	6.47e-01	-6.49e-01	-6.03e-01
Vds:	8.12e-01	1.65e-01	4.53e-01	-4.27e-01	-1.76e-01
Vbs:	0.00e+00	0.00e+00	-1.65e-01	1.76e-01	0.00e+00
Vth:	5.08e-01	5.10e-01	5.54e-01	-5.58e-01	-5.03e-01
Vdsat:	2.00e-01	1.13e-01	1.07e-01	-1.24e-01	-1.28e-01
Gm:	4.46e-03	2.12e-03	2.33e-03	2.57e-03	2.38e-03
Gds:	1.54e-04	2.71e-04	7.63e-05	7.84e-05	2.22e-04
Gmb:	1.26e-03	6.11e-04	5.96e-04	7.94e-04	8.07e-04
Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgsov:	4.02e-15	4.02e-15	4.02e-15	1.90e-14	1.90e-14
Cgdov:	4.02e-15	4.02e-15	4.02e-15	1.90e-14	1.90e-14
Cgbov:	3.18e-17	3.18e-17	3.18e-17	4.83e-17	4.83e-17
dQgdVgb:	2.10e-14	2.02e-14	1.98e-14	9.98e-14	1.01e-13
dQgdVdb:	-4.01e-15	-4.12e-15	-4.00e-15	-1.88e-14	-1.93e-14
dQgdVsb:	-1.55e-14	-1.43e-14	-1.39e-14	-7.51e-14	-7.63e-14
dQddVgb:	-4.04e-15	-4.39e-15	-4.06e-15	-1.92e-14	-2.05e-14
dQddVdb:	4.03e-15	4.36e-15	4.04e-15	1.91e-14	2.03e-14
dQddVsb:	1.26e-17	1.45e-16	3.01e-17	1.83e-16	7.83e-16
dQbdVgb:	-2.62e-15	-2.47e-15	-2.53e-15	-1.32e-14	-1.31e-14
dQbdVdb:	6.22e-19	-1.05e-16	1.59e-18	4.35e-18	-3.88e-16
dQbdVsb:	-1.90e-15	-2.08e-15	-1.75e-15	-7.38e-15	-9.18e-15
Name:	mp3				
Model:	pch.3				
Id:	-1.47e-03				
Vgs:	-8.25e-01				
Vds:	-8.25e-01				
Vbs:	0.00e+00				
Vth:	-5.00e-01				
Vdsat:	-2.74e-01				
Gm:	7.38e-03				
Gds:	2.78e-04				

Fig 3. Spice Error Log Window for V_{gt} Check

The details are summarized as follows in Table 2:

	V_{gs}	V_{th}	V_{gt}
M_{n1}	0.618 V	0.51 V	108 mV
M_{n2}	0.647 V	0.554 V	0.93 mV
M_{p1}	-0.603 V	-0.503 V	100 mV
M_{p2}	-0.649 V	-0.558 V	91 mV

Table 2. Sizing parameters of transistors for V_{gt} Requirements

From Table 2. the obtained value of V_{gt} for transistors M_{n1} , M_{n2} and M_{p1} , M_{p2} are found to be approximately equal to $100mV$ which satisfies our requirements.

(Part b)

In the problem statement, the ratio of $I_{b,ncas}$ and $I_{b,pcas}$ should be 4 times larger than $I_{b,main}$ for an ideal condition. But in a practical scenario, the value of **ratio_n** and **ratio_p** should be chosen such that the condition where transistors M_{n1} and M_{p1} are on the edge of saturation with a margin of approximately $50mV$ is satisfied.

To satisfy this condition of edge of saturation, the ratios of $I_{b,ncas}$ and $I_{b,pcas}$ are tweaked such that: -

1. $I_{b,ncas} = ratio_n * I_{b,main}$ such that **ratio_n equals to 5.**
2. $I_{b,pcas} = ratio_p * I_{b,main}$ such that **ratio_p equals to 7.35.**

Choosing these values based on the logic as given below: -

From the Fig 4. as we can see that both M_{n1} and M_{p1} are on the verge of saturation as evident from Table 3.

	V_{ds}	V_{dsat}	V_{margin}	Ratio
M_{n1}	0.165 V	0.113 V	52 mV	5
M_{p2}	0.128 V	0.176 V	48 mV	7.35

Table 3. Ratio Parameters for saturation margin

SPICE Error Log: C:\Users\harsh\Downloads\HW1_Sim1.log					
Semiconductor Device Operating Points:					
--- BSIM3 MOSFETS ---					
Name:	mn4	mn5	mn3	mn1	mn2
Model:	nch.3	nch.3	nch.3	nch.3	nch.3
Id:	2.05e-04	2.05e-04	1.00e-03	2.00e-04	2.00e-04
Vgs:	6.18e-01	6.27e-01	8.12e-01	6.18e-01	6.47e-01
Vds:	1.85e-01	1.62e+00	8.12e-01	1.65e-01	4.53e-01
Vbs:	0.00e+00	-1.85e-01	0.00e+00	0.00e+00	-1.65e-01
Vth:	5.10e-01	5.55e-01	5.08e-01	5.10e-01	5.54e-01
Vdsat:	1.13e-01	9.83e-02	2.00e-01	1.13e-01	1.07e-01
Gm:	2.18e-03	2.45e-03	4.46e-03	2.12e-03	2.33e-03
Gds:	2.25e-04	4.76e-05	1.54e-04	2.71e-04	7.63e-05
Gmb:	6.30e-04	5.94e-04	1.26e-03	6.11e-04	5.96e-04
Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgsov:	4.02e-15	4.02e-15	4.02e-15	4.02e-15	4.02e-15
Cgdov:	4.02e-15	4.02e-15	4.02e-15	4.02e-15	4.02e-15
Cgbov:	3.18e-17	3.18e-17	3.18e-17	3.18e-17	3.18e-17
dQgdVgb:	2.02e-14	1.93e-14	2.10e-14	2.02e-14	1.98e-14
dQgdVdb:	-4.09e-15	-3.99e-15	-4.01e-15	-4.12e-15	-4.00e-15
dQgdVsb:	-1.43e-14	-1.33e-14	-1.55e-14	-1.43e-14	-1.39e-14
dQddVgb:	-4.31e-15	-4.03e-15	-4.04e-15	-4.39e-15	-4.06e-15
dQddVdb:	4.26e-15	4.02e-15	4.03e-15	4.36e-15	4.04e-15
dQddVsb:	1.31e-16	2.44e-18	1.26e-17	1.45e-16	3.01e-17
dQbdVgb:	-2.50e-15	-2.58e-15	-2.62e-15	-2.47e-15	-2.53e-15
dQbdVdb:	-7.38e-17	5.47e-18	6.22e-19	-1.05e-16	1.59e-18
dQbdVsb:	-2.07e-15	-1.66e-15	-1.90e-15	-2.08e-15	-1.75e-15
Name:	mp5	mp4	mp2	mp1	mp3
Model:	pch.3	pch.3	pch.3	pch.3	pch.3
Id:	1.74e-18	-5.60e-19	-2.00e-04	-2.00e-04	-1.47e-03
Vgs:	-8.25e-01	-6.03e-01	-6.49e-01	-6.03e-01	-8.25e-01
Vds:	2.22e-16	-2.22e-16	-4.27e-01	-1.76e-01	-8.25e-01
Vbs:	2.22e-16	0.00e+00	1.76e-01	0.00e+00	0.00e+00
Vth:	-5.03e-01	-5.03e-01	-5.58e-01	-5.03e-01	-5.00e-01
Vdsat:	-2.72e-01	-1.28e-01	-1.24e-01	-1.28e-01	-2.74e-01
Gm:	5.58e-18	4.17e-18	2.57e-03	2.38e-03	7.38e-03
Gds:	7.55e-03	2.67e-03	7.84e-05	2.22e-04	2.78e-04
Gmb:	2.05e-18	1.46e-18	7.94e-04	8.07e-04	2.49e-03
Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00

Fig 4. Spice Error Log Info for saturation margin of Mn1 and Mp1

Using the details in Fig 4. we can see that the current through transistor M_{p3} is $I_{b,pcas}$ with value equal to 147 μA . Similarly the current flowing the transistor M_{n3} is $I_{b,ncas}$ with value equal to 100 μA . This pretty much shows that the ratios in a practical scenario in comparison to $I_{b,main}$ is definitely not 4. So the values for $ratio_p$ and $ratio_n$ are tweaked based on the following logic:

$$\text{We know that } -V_{ov} = \sqrt{\frac{2I_d}{\mu C_{ox} \left(\frac{W}{L}\right)}}$$

So the overdrive voltage value of transistor M_{n3} is larger than twice the value of overdrive voltage value of transistor M_{n1} and M_{n1} . Then we follow this same equation to get the relation between drain-source voltages M_{n1} , M_{n2} and M_{p1} , M_{p2} and their overdrive voltage values such that: -

$$\diamond V_{ds,Mn1} > V_{ov,Mn1}$$

instead of

$$\diamond V_{ds,Mn1} = V_{ov,Mn1}$$

Here, we can adjust the saturation margin voltage of transistors M_{n1} , M_{n2} and M_{p1} , M_{p2} to 50mV by adjusting the ratio_n and ratio_p such the values of $I_{b,pcas}$ and $I_{b,ncas}$ increases.

Since the value of M factor is larger for PMOS transistor (where $\{W/L = 58\}$) in comparison to M factor for NMOS transistors (where $\{W/L = 11\}$), the value of $I_{b,pcas}$ is larger than $I_{b,ncas}$. From Fig 4. we get that $I_{b,pcas} = 147 \text{ uA}$ and $I_{b,ncas} = 100 \text{ uA}$, which fulfils the requirement of satisfying the requirement of voltage margin to be approximately 50 mV.

(Part c)

Keeping the ratios for $I_{b,pcas}$ and $I_{b,ncas}$ the same, we reduce the value of $I_{b,main}$ by 2 times, we get: -

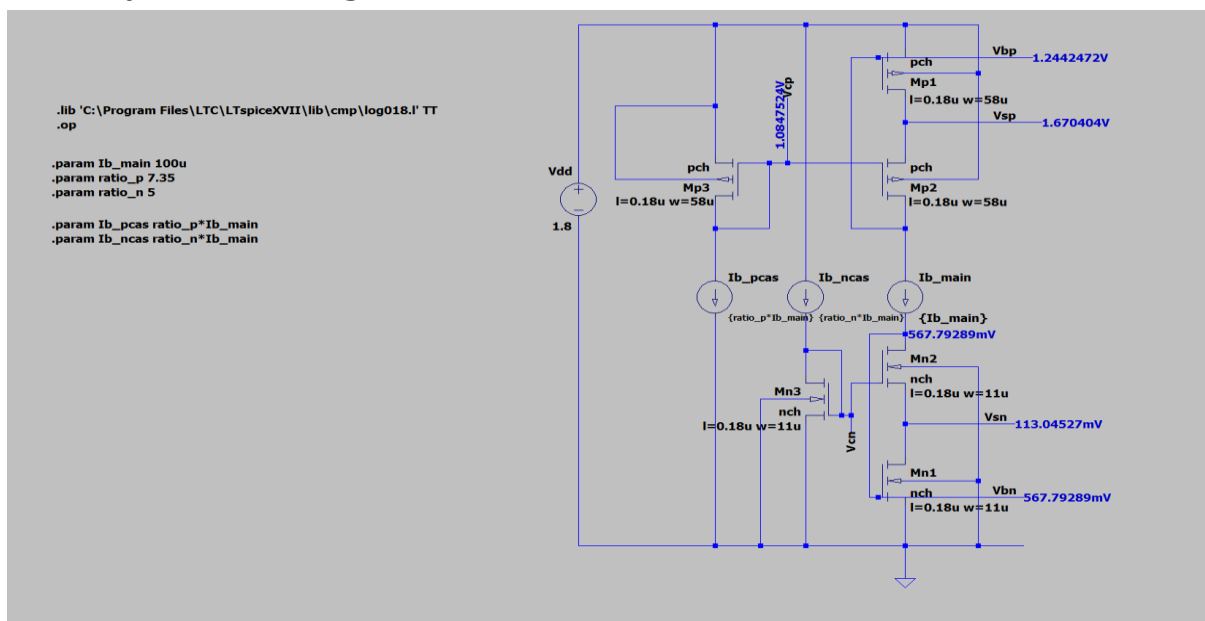


Fig 5. Schematic Diagram at $(I_{b,main}/2)$

SPICE Error Log: C:\Users\harsh\Downloads\HW1_Sim1.log					
Semiconductor Device Operating Points:					
--- BSIM3 MOSFETS ---					
Name:	mn3	mn1	mn2	mp2	mp1
Model:	nch.3	nch.3	nch.3	pch.3	pch.3
Id:	5.00e-04	1.00e-04	1.00e-04	-1.00e-04	-1.00e-04
Vgs:	6.94e-01	5.68e-01	5.81e-01	-5.86e-01	-5.56e-01
Vds:	6.94e-01	1.13e-01	4.55e-01	-4.26e-01	-1.30e-01
Vbs:	0.00e+00	0.00e+00	-1.13e-01	1.30e-01	0.00e+00
Vth:	5.09e-01	5.10e-01	5.41e-01	-5.44e-01	-5.03e-01
Vdsat:	1.49e-01	8.99e-02	8.35e-02	-9.50e-02	-9.98e-02
Gm:	3.64e-03	1.34e-03	1.49e-03	1.58e-03	1.46e-03
Gds:	1.08e-04	2.33e-04	4.24e-05	4.04e-05	1.62e-04
Gmb:	1.03e-03	3.88e-04	3.95e-04	4.97e-04	4.94e-04
Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgsov:	4.02e-15	4.02e-15	4.02e-15	1.90e-14	1.90e-14
Cgdov:	4.02e-15	4.02e-15	4.02e-15	1.90e-14	1.90e-14
Cgbov:	3.18e-17	3.18e-17	3.18e-17	4.83e-17	4.83e-17
dQgdVgb:	2.07e-14	1.91e-14	1.82e-14	9.11e-14	9.44e-14
dQgdVdb:	-4.01e-15	-4.14e-15	-4.00e-15	-1.89e-14	-1.93e-14
dQgdVsb:	-1.50e-14	-1.28e-14	-1.18e-14	-6.36e-14	-6.70e-14
dQddVgb:	-4.04e-15	-4.49e-15	-4.05e-15	-1.91e-14	-2.07e-14
dQddVdb:	4.03e-15	4.42e-15	4.03e-15	1.90e-14	2.03e-14
dQddVsb:	1.60e-17	2.01e-16	2.30e-17	1.43e-16	9.95e-16
dQbdVgb:	-2.58e-15	-2.60e-15	-2.79e-15	-1.41e-14	-1.37e-14
dQbdVdb:	1.40e-18	-1.27e-16	2.54e-18	1.28e-17	-4.02e-16
dQbdVsb:	-2.00e-15	-1.92e-15	-1.56e-15	-6.84e-15	-8.50e-15
Name:	mp3				
Model:	pch.3				
Id:	-7.35e-04				
Vgs:	-7.15e-01				
Vds:	-7.15e-01				
Vbs:	0.00e+00				
Vth:	-5.01e-01				
Vdsat:	-2.03e-01				
Gm:	5.48e-03				
Gds:	1.66e-04				
Gmb:	1.84e-03				

Fig 6. Error Log details at ($I_{b,main}/2$)

Keeping the ratios for $I_{b,pcas}$ and $I_{b,ncas}$ the same, we reduce the value of $I_{b,main}$ by 2 times, we get: -

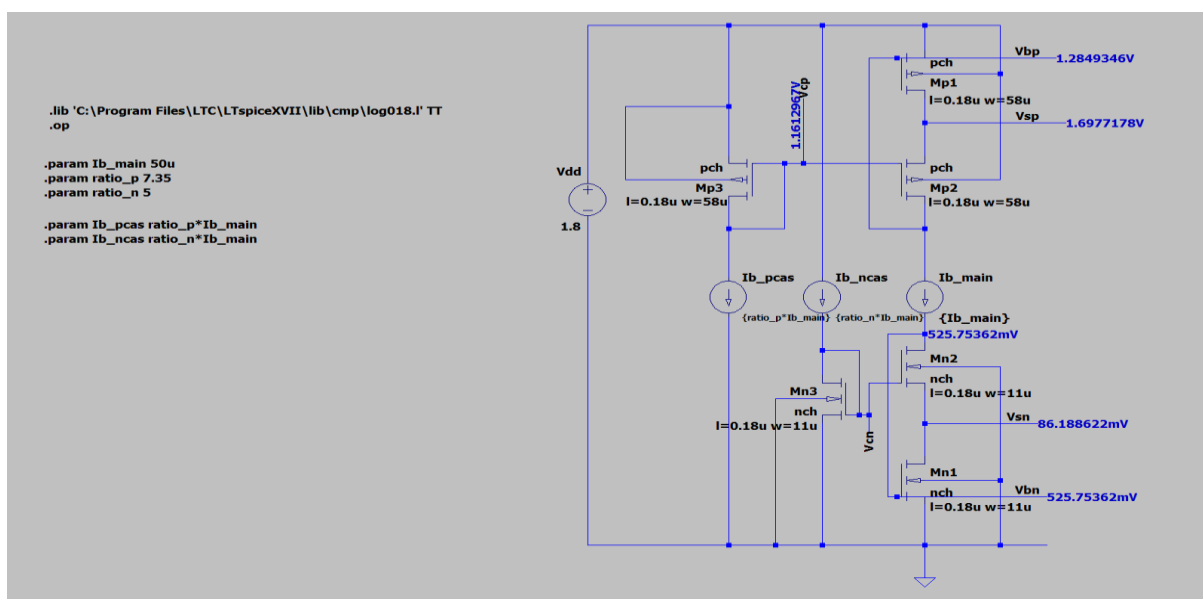


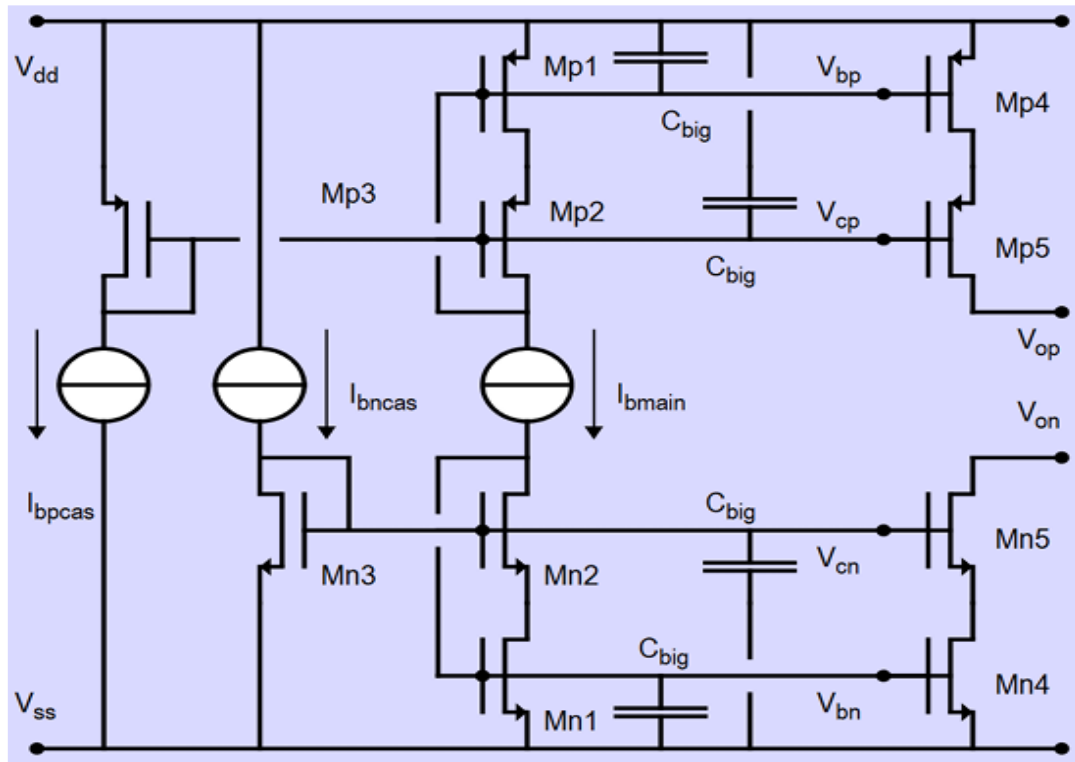
Fig 7. Schematic Diagram at ($I_{b,main}/4$)

SPICE Error Log: C:\Users\harsh\Downloads\HW1_Sim1.log					
Name:	mn3	mn1	mn2	mp2	mp1
Model:	nch.3	nch.3	nch.3	pch.3	pch.3
Id:	2.50e-04	5.00e-05	5.00e-05	-5.00e-05	-5.00e-05
Vgs:	6.18e-01	5.26e-01	5.32e-01	-5.36e-01	-5.15e-01
Vds:	6.18e-01	8.62e-02	4.40e-01	-4.13e-01	-1.02e-01
Vbs:	0.00e+00	0.00e+00	-8.62e-02	1.02e-01	0.00e+00
Vth:	5.09e-01	5.10e-01	5.34e-01	-5.35e-01	-5.03e-01
Vdsat:	1.13e-01	7.28e-02	6.73e-02	-7.57e-02	-7.96e-02
Gm:	2.66e-03	8.01e-04	8.78e-04	9.05e-04	8.48e-04
Gds:	7.01e-05	1.68e-04	2.38e-05	2.15e-05	1.07e-04
Gmb:	7.55e-04	2.33e-04	2.38e-04	2.89e-04	2.88e-04
Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgsov:	4.02e-15	4.02e-15	4.02e-15	1.90e-14	1.90e-14
Cgdov:	4.02e-15	4.02e-15	4.02e-15	1.90e-14	1.90e-14
Cgbov:	3.18e-17	3.18e-17	3.18e-17	4.83e-17	4.83e-17
dQgdVgb:	2.01e-14	1.71e-14	1.59e-14	7.89e-14	8.33e-14
dQgdVdb:	-4.01e-15	-4.12e-15	-4.01e-15	-1.89e-14	-1.93e-14
dQgdVsb:	-1.42e-14	-1.02e-14	-8.81e-15	-4.72e-14	-5.20e-14
dQddVgb:	-4.04e-15	-4.41e-15	-4.04e-15	-1.91e-14	-2.04e-14
dQddVdb:	4.03e-15	4.34e-15	4.03e-15	1.90e-14	2.00e-14
dQddVsb:	1.81e-17	1.87e-16	1.47e-17	9.36e-17	8.82e-16
dQbdVgb:	-2.59e-15	-3.00e-15	-3.25e-15	-1.59e-14	-1.54e-14
dQbdVdb:	2.13e-18	-9.86e-17	1.87e-18	1.09e-17	-3.00e-16
dQbdVsb:	-2.03e-15	-1.46e-15	-1.06e-15	-4.89e-15	-6.43e-15
Name:	mp3				
Model:	pch.3				
Id:	-3.68e-04				
Vgs:	-6.39e-01				
Vds:	-6.39e-01				
Vbs:	0.00e+00				
Vth:	-5.01e-01				
Vdsat:	-1.52e-01				
Gm:	3.82e-03				
Gds:	9.69e-05				
Gmb:	1.28e-03				
Cbd:	0.00e+00				
Cbs:	0.00e+00				
Cgsov:	1.90e-14				

Fig 8. Error Log details at ($I_{b,main}/4$)

From the details of Fig 6 and Fig 8. It is quite evident that the values of V_{gs} and V_{dsat} decreases when the value of $I_{b,main}$ is reduced 2 times and 4 times respectively. Due to biased current flowing the transistors M_{p1} and M_{n1} , the circuit will fall into saturation region quickly as V_{ds} further reduced from the original value of $I_{b,main}$. With this, the edge of the saturation become higher than 50 mV in both the cases.

Simulation 2



Given Details: -

- Using the schematic in the first simulation, add transistor M_{p4} , M_{p5} and M_{n4} , M_{n5} as shown in the circuit above and then add the capacitance value to be 1F.

Required Targets to be achieved in Simulation 2: -

- Sweep both V_{op} and V_{on} individually and measure the current in each branch. Make a plot of $I(V_{op})$ versus V_{op} and $I(V_{on})$ versus V_{on} .
- Determine for both branches what the minimum voltage is across the 2 transistors (so either M_{n4} and M_{n5} or M_{p4} and M_{p5}) to keep the devices saturated.
- Plot the output impedance of each branch in the same figure.

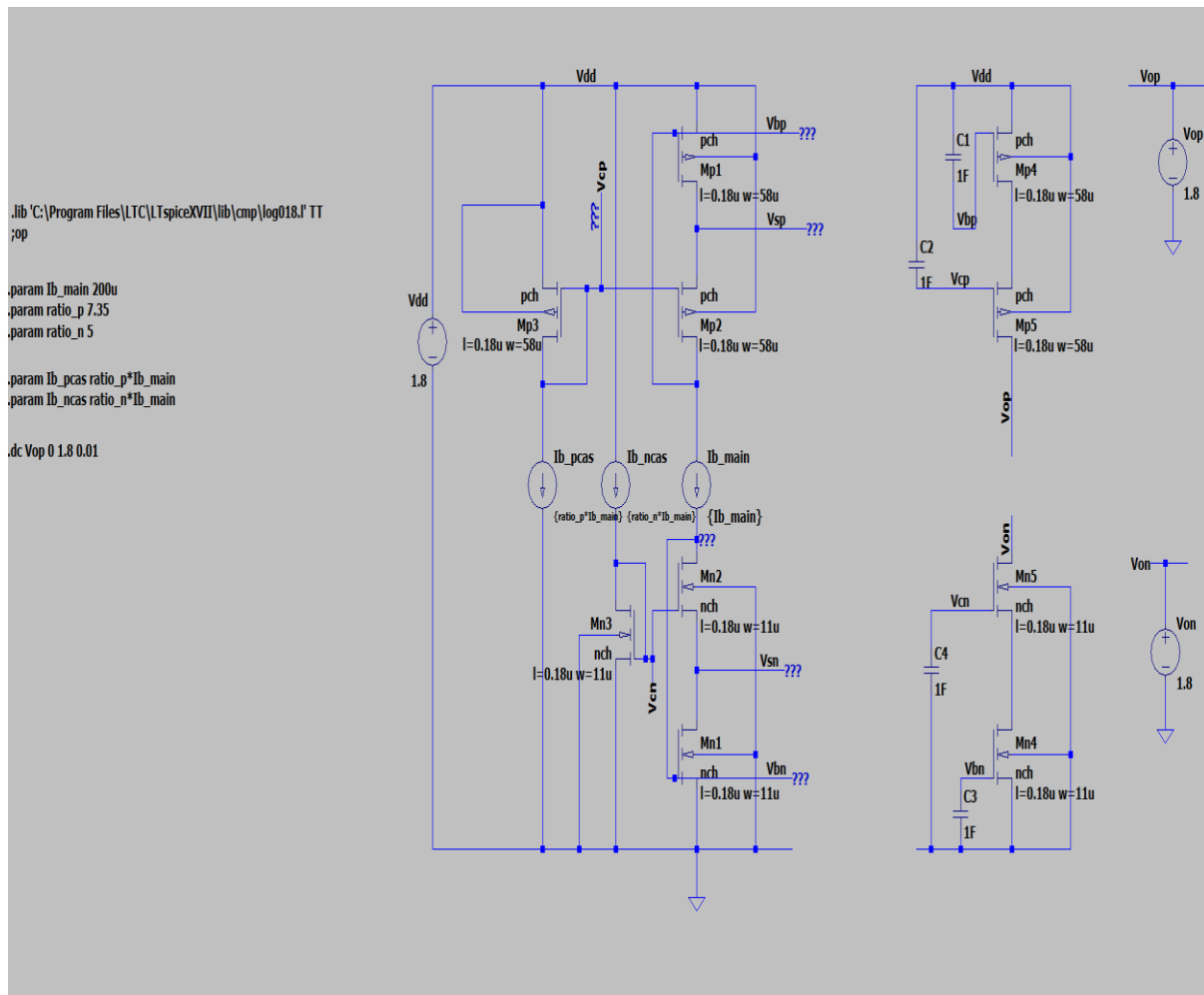


Fig 10. Schematic Diagram for Simulation2 Circuit

From the circuit in Fig 10, the simulation results are obtained in Fig. 11 where the minimum and maximum output voltage values have been obtained as **184.79 V and 204.27 V respectively**. The value is obtained through checking the operating point file (attached with the report in submission). The voltage equals $V_{ds(Mn4)} + V_{dsat(Mn5)}$. The same logic also applies to the PMOS branch.

Apart from this, the output impedances for two branches of Vop and Von is also plotted in Fig.11, with light blue line indicating the results for PMOS branch and red line indicating the results for NMOS branch.

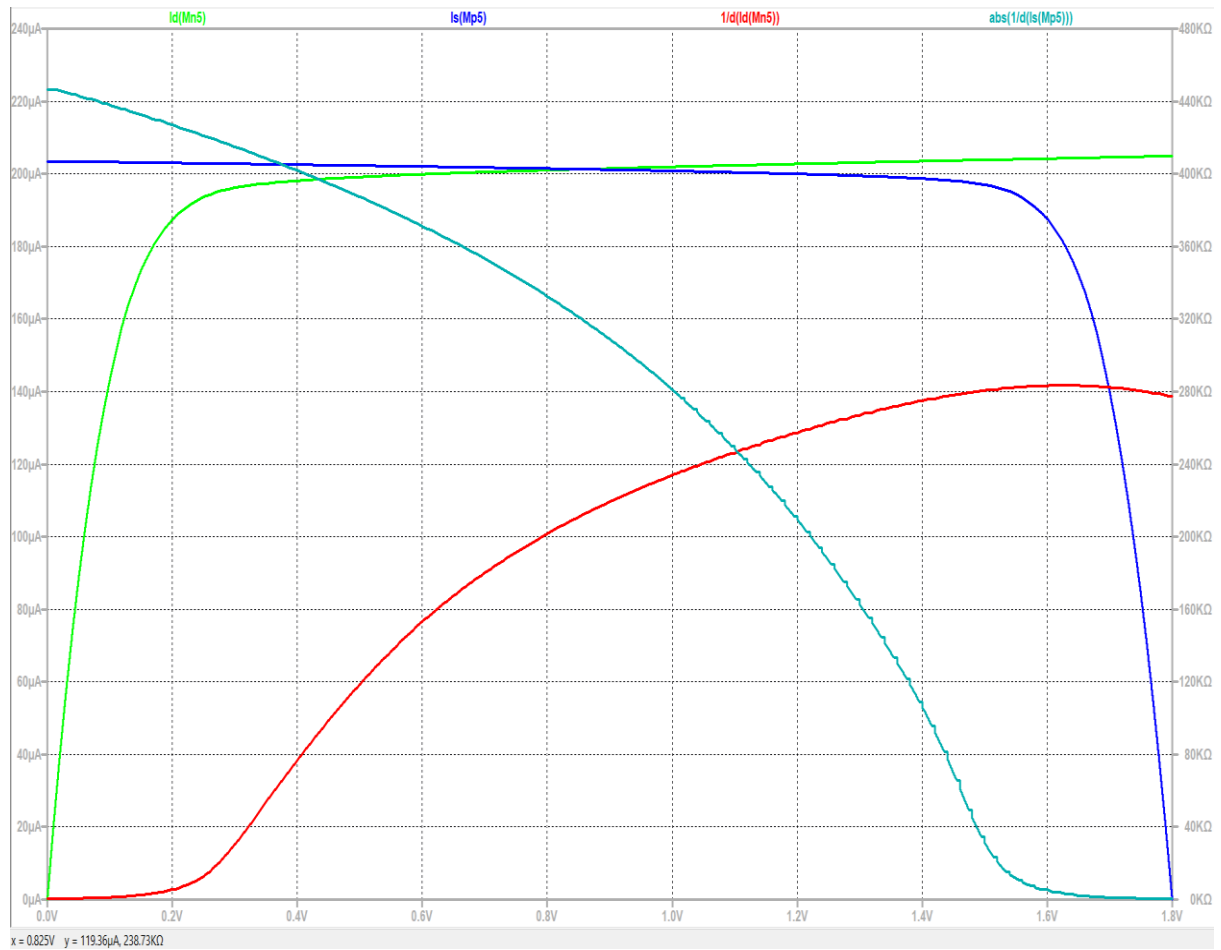


Fig 11. Simulation Results for $I(V_{op})$ v/s V_{op} and $I(V_{on})$ v/s V_{on} and Out Impedance